

Basic Principles of Medicine 2
Module : Endocrinology &
Reproduction
Lecture No: 8

Diabetes and Hypoglycemia

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Acknowledgement: Dr. Mohamed Idries

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Objectives: Required reading: Endocrine Pancreas: Insulin and Glucagon

- Required reading before lecture 'Diabetes mellitus and hypoglycemia'
-  <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>; Chapter 23: Metabolic effects of insulin and glucagon
- [23. Metabolic Effects of Insulin and Glucagon | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

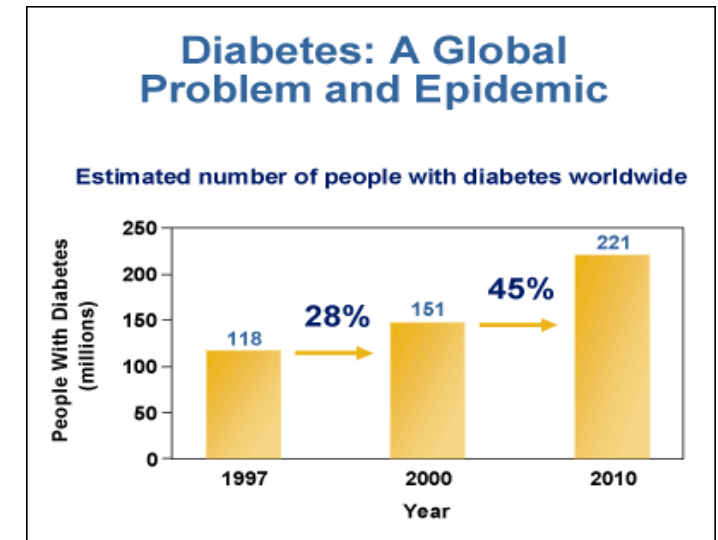
SOM.MK.II.BPM2.1.ER.2.BCHM.1047	Identify the major hormones secreted from the endocrine pancreas, their cells of origin, and their chemical nature
SOM.MK.II.BPM2.1.ER.2.BCHM.1048	Describe the major stimulus for synthesis, secretion and regulation of insulin
SOM.MK.II.BPM2.1.ER.2.BCHM.1049	Describe the hormonal changes after a carbohydrate-rich meal and the function of these hormones.
SOM.MK.II.BPM2.1.ER.2.BCHM.1050	Explain insulin formation and secretion by the pancreatic beta- cells. Highlight the significance of C-peptide.
SOM.MK.II.BPM2.1.ER.2.BCHM.1052	Describe the regulation of glucagon secretion.
SOM.MK.II.BPM2.1.ER.2.BCHM.1053	Describe the principal actions of glucagon on target tissues including intracellular signal transduction.

Objectives: Diabetes Mellitus and Hypoglycemia

SOM.MK.II.BPM2.1.ER.2.BCHM.1034	Distinguish between the types of diabetes: type 1 and type 2 (differences in insulin secretion) and secondary diabetes mellitus.
SOM.MK.II.BPM2.1.ER.2.BCHM.1035	Explain the role of free fatty acids, leptin, adiponectin, incretins and proinflammatory cytokines in development of insulin resistance in obesity and type 2 diabetes mellitus
SOM.MK.II.BPM2.1.ER.2.BCHM.1036	Correlate presenting features of type 1 and type 2 diabetes mellitus and basis for clinical features
SOM.MK.II.BPM2.1.ER.2.BCHM.1037	Evaluate changes in serum lipids (dyslipidemia) in diabetes mellitus and correlate to occurrence of macrovascular changes.
SOM.MK.II.BPM2.1.ER.2.BCHM.1038	Define criteria for diagnosis of diabetes mellitus: Fasting plasma glucose, 2-hour plasma glucose following oral glucose challenge, HbA1c, role of OGTT
SOM.MK.II.BPM2.1.ER.2.BCHM.1039	Develop the concept of medical nutrition therapy; Highlight the significance of dietary carbohydrates, artificial sweeteners, Glycemic index, Dietary lipids and Mediterranean diet, Dietary fiber in diabetics
SOM.MK.II.BPM2.1.ER.2.BCHM.1040	Evaluate significance of exercise and weight management in management of patients with prediabetes, metabolic syndrome and diabetes mellitus
SOM.MK.II.BPM2.1.ER.2.BCHM.1041	Identify disease states caused by: a) over secretion, b) under secretion of insulin, or c) decreased sensitivity to insulin, D) hypoglycemia
SOM.MK.II.BPM2.1.ER.2.BCHM.1043	Summarize the release of insulin-counterregulatory hormones like glucagon, epinephrine and cortisol.
SOM.MK.II.BPM2.1.ER.2.BCHM.1044	Describe the adrenergic and neuroglycopenic symptoms in patients with hypoglycemia
SOM.MK.II.BPM2.1.ER.2.BCHM.1045	Differentiate hypoglycemia due to a. Injection of insulin. b. Reactive (postprandial) hypoglycemia. c. Alcohol induced e. Insulinoma. f. Factitious (Insulin injection, sulfonylurea).
SOM.MK.II.BPM2.1.ER.2.BCHM.1046	Justify the significance of estimation of serum insulin, C-peptide, prosinsulin in the differential diagnosis of various types of hypoglycemia

Diabetes Mellitus

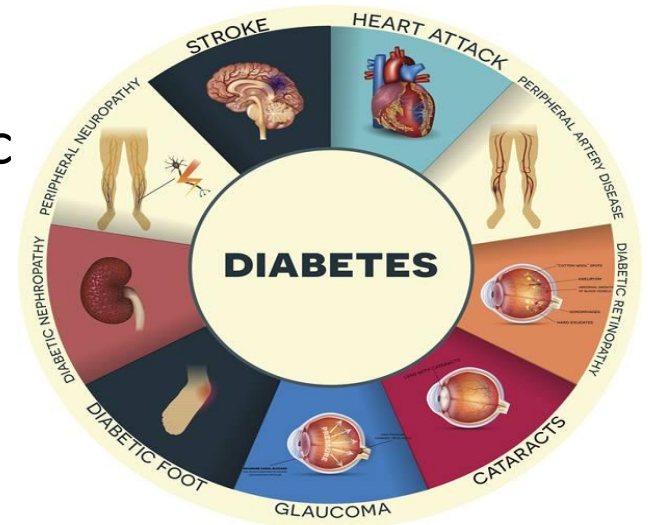
- Heterogenous group of common metabolic disorders sharing the phenotype of hyperglycemia.
- Associated with either:
 1. Relative/absolute **impaired insulin secretion**.
 2. Varying peripheral **resistance to insulin action**.
- Etiologic classification:
 1. Type **1** diabetes mellitus (**β -cell destruction**)
 2. Type **2** diabetes mellitus (**insulin resistance**)
 3. Genetic defects of β -cell function or insulin action
 4. Secondary: endocrinopathies, exocrine pancreas disease, drug induced, infections, etc.
 5. Gestational diabetes



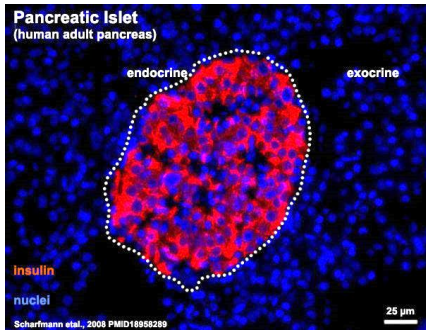
[Diabetes Mellitus | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

Epidemiology of Diabetes mellitus

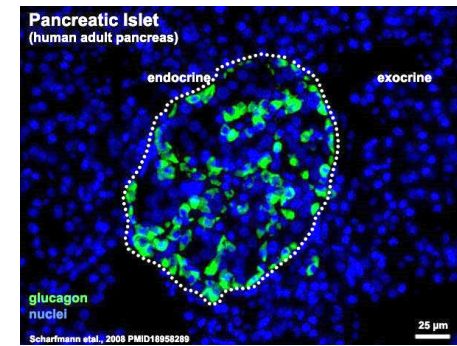
- Affects multiple organ systems.
- Leading cause of end-stage renal disease (ESRD), nontraumatic lower limb amputation, and adult blindness.
- Leads to increased risk for cardiovascular disease, main cause of morbidity and mortality.
- Worldwide prevalence increased in past two decades (concomitant increase in obesity: 'Diabesity').
- Type 2 DM accounts for 90% of cases.
- Type 1 DM accounts for 5 – 10% of cases.



Hormone-producing cell types in the islets



Insulin



Glucagon

- **Beta cells (70%):** Release **insulin** which lowers blood glucose level
 - Stimulates synthesis of glycogen, protein and fatty acids.
- **Alpha cells (20%):** Release **glucagon** which increases the blood glucose level
 - Stimulates glycogen breakdown and gluconeogenesis in liver and kidney
- **Delta cells (10%):** Release **somatostatin** which inhibits release of both insulin and glucagon in the islets (paracrine).
- **F-(PP) cells (5%):** Release **pancreatic polypeptide hormone** which inhibits pancreatic exocrine secretions. Also, may act as a satiety hormone.

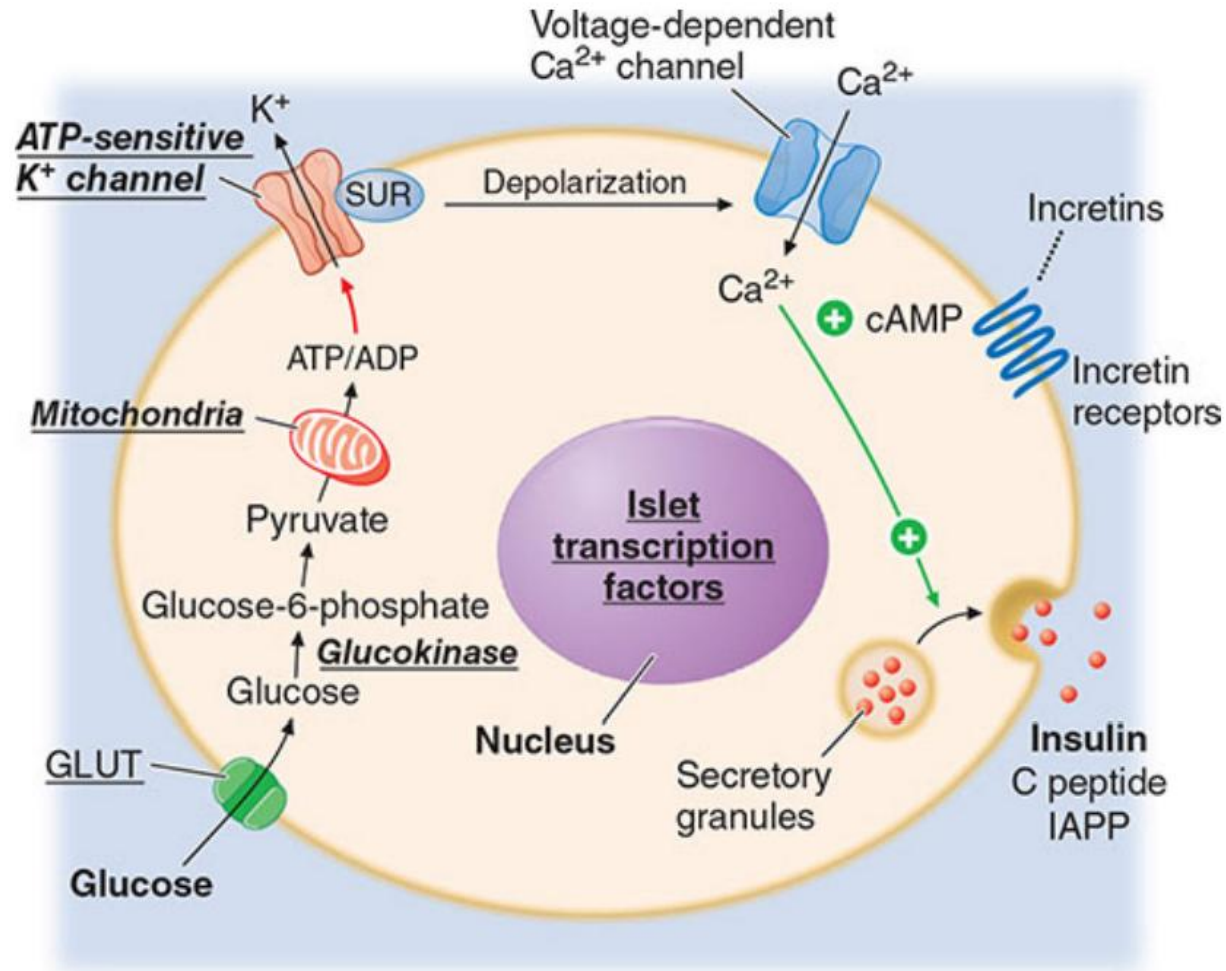
Mechanism of Insulin secretion

1. Glucose entry via GLUT-2 into pancreatic β -cells $\rightarrow$ Glucokinase phosphorylation glucose $\rightarrow$ glycolysis $\rightarrow$ ATP production

2. ATP inhibits & closes ATP-sensitive K^+ channel

3. Cell membrane depolarization $\rightarrow$ opens voltage-gated calcium channel $\rightarrow$ calcium influx

4. Release of insulin and C-peptide at a 1:1 ratio from granules



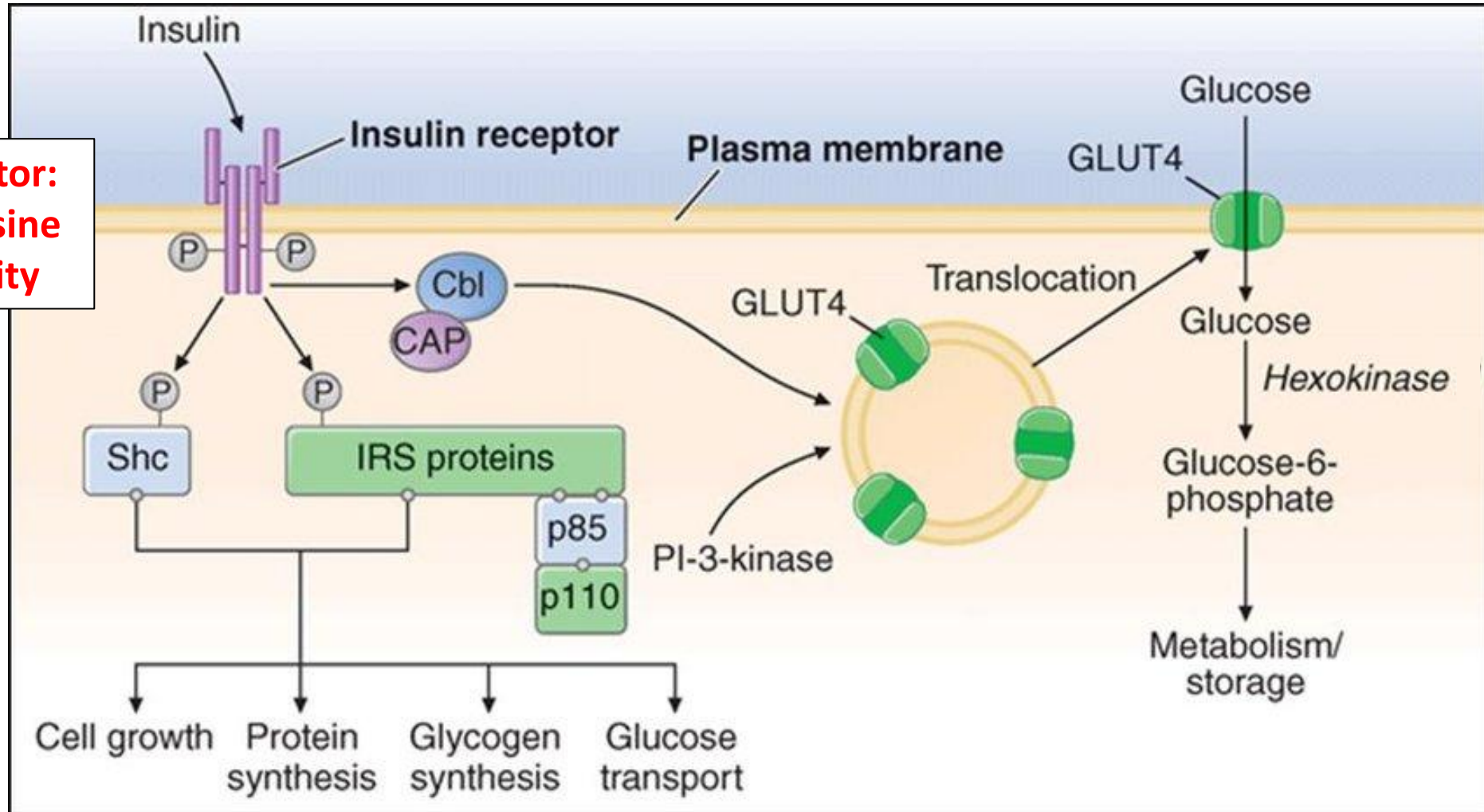
Actions of Insulin

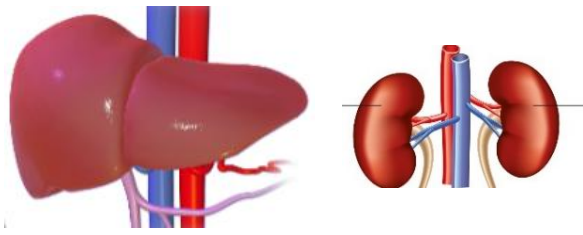
- Insulin is secreted into portal venous system, acts as an **anabolic hormone**.
- Binds to insulin receptor stimulating intrinsic tyrosine kinase activity.

Liver	Adipose Tissue	Skeletal Muscles
Stimulation: - Glucose uptake, utilization. - Glycogen synthesis. - Fatty acids, TAGs, and cholesterol synthesis; and VLDL release Inhibition: - Gluconeogenesis. - Glycogen breakdown	Stimulation: - GLUT-4 uptake of glucose and glycolysis for glycerol synthesis of TAGs. - Lipoprotein lipase (adipose tissue capillaries) → fatty acids uptake into adipocytes. - Synthesis & storage of TAGs. Inhibition: - Decreased lipolysis.	Stimulation: - GLUT-4 mediated uptake of glucose and glycolysis. - Glycogen synthesis. - Amino acid uptake → Protein synthesis. - Na⁺/K⁺-ATPase synthesis.

Mechanism of insulin action on target cells

**Insulin receptor:
Intrinsic tyrosine
kinase activity**



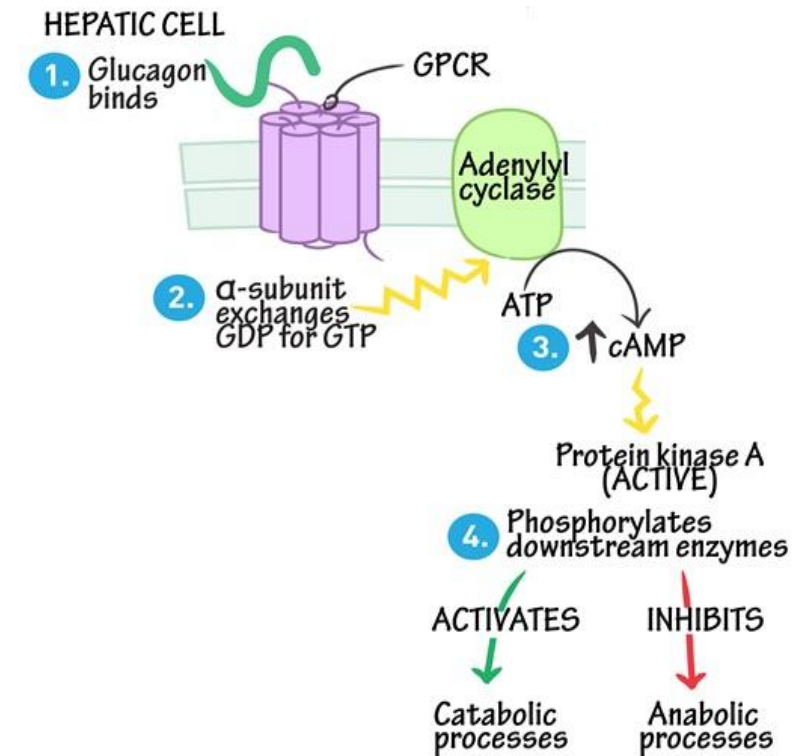


Glucagon

KEY POINTS FROM REQUIRED READING

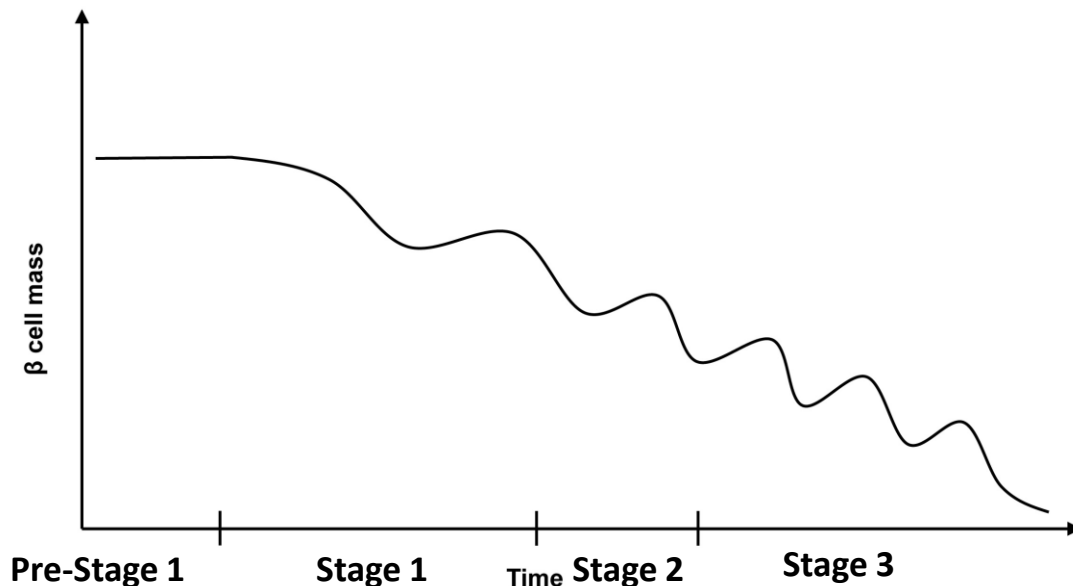
Glucagon binds to G protein-coupled receptors (**GPCR**) and acts via **cAMP-messenger system**

- **Activators** of glucagon secretion:
 - Amino acids; Low blood glucose
 - Epinephrine and norepinephrine
 - Cortisol
 - Neural input: sympathetic stimulation during stress
- **Inhibitors** of glucagon secretion:
 - Increased blood glucose
 - Insulin
 - Somatostatin
- Action and target organs:
 - Target organs: **Hepatocytes** and **renal cortex cells**
 - Activates glycogenolysis and gluconeogenesis
 - Increases plasma glucose level



Type 1 Diabetes Mellitus

- **Autoimmune destruction** of pancreatic β -cells; In genetically susceptible individuals.
- Triggered by environmental factors (e.g., viral infections, toxins, dietary components)
- Progressive destruction of pancreatic β -cells over months or years → **Very low/absent insulin and C-peptide** → Abrupt onset of diabetes once >90% of β -cells are destroyed.
- Life-threatening complication: **ketoacidosis**; Prevented by **life-long insulin**.



- Age of onset: Adolescence or young adults
- Progression of type 1 diabetes mellitus:
 - **Pre-stage 1:** Genetic susceptibility
 - **Stage 1:** Environmental factors trigger autoimmune response, euglycemia
 - **Stage 2:** Dysglycemia and presence of islet autoantibodies (**glutamic acid decarboxylase antibodies**)
 - **Stage 3:** Diabetes mellitus

Type 2 Diabetes Mellitus

- Multifactorial disorder
- Risk factors:
 1. Family history
 2. Abdominal obesity (Metabolic syndrome)
 3. Ethnicity
 4. Age
 5. Sedentary lifestyle
- Ketoacidosis **not as common** as in type 1 DM (enough insulin to prevent ketoacidosis but not enough insulin to prevent hyperglycemia)
- Management with **oral hypoglycemic drugs initially** and with **insulin in later stages**

Insulin resistance in target tissue (liver, skeletal muscle, adipose tissue)



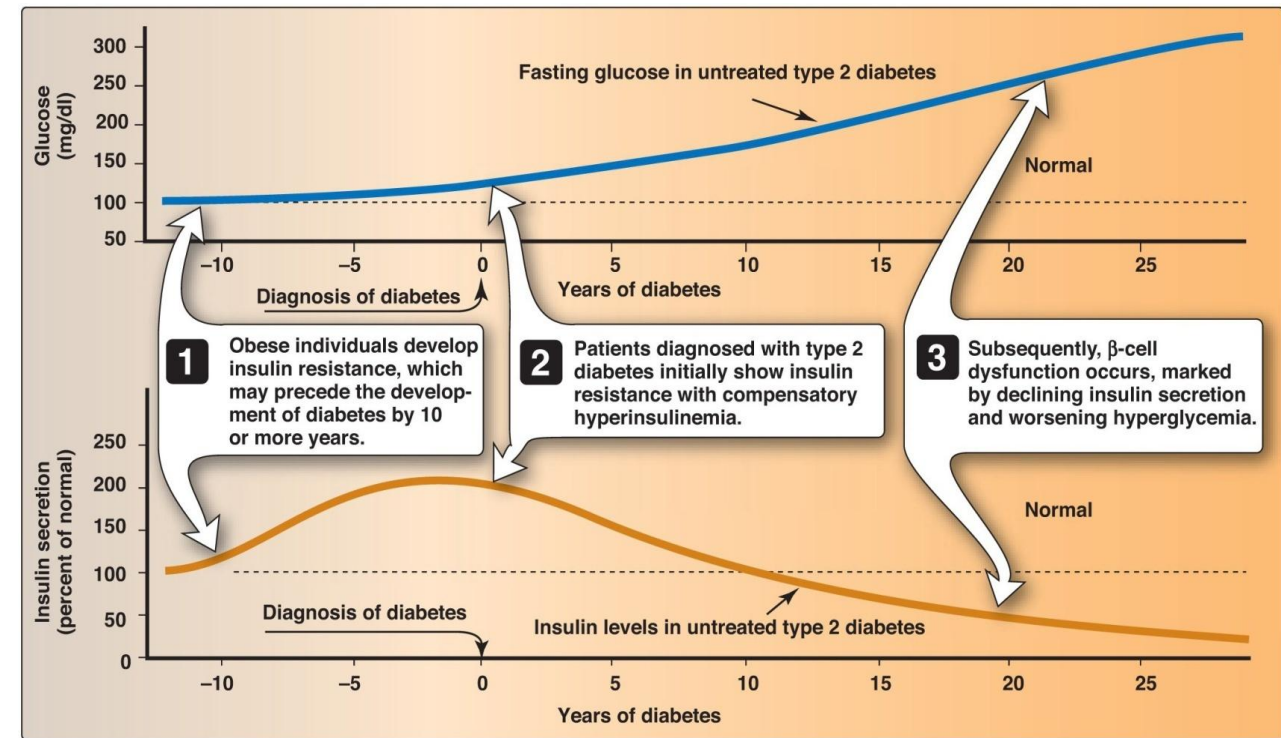
Compensatory increase insulin production by β -cells to overcome resistance (**initial elevated levels of insulin**)



Exhaustion of β -cells $\rightarrow$ loss of adaptive capacity $\rightarrow$ **relative insulin deficiency** (**later inadequate insulin** level in face of high glucose level)

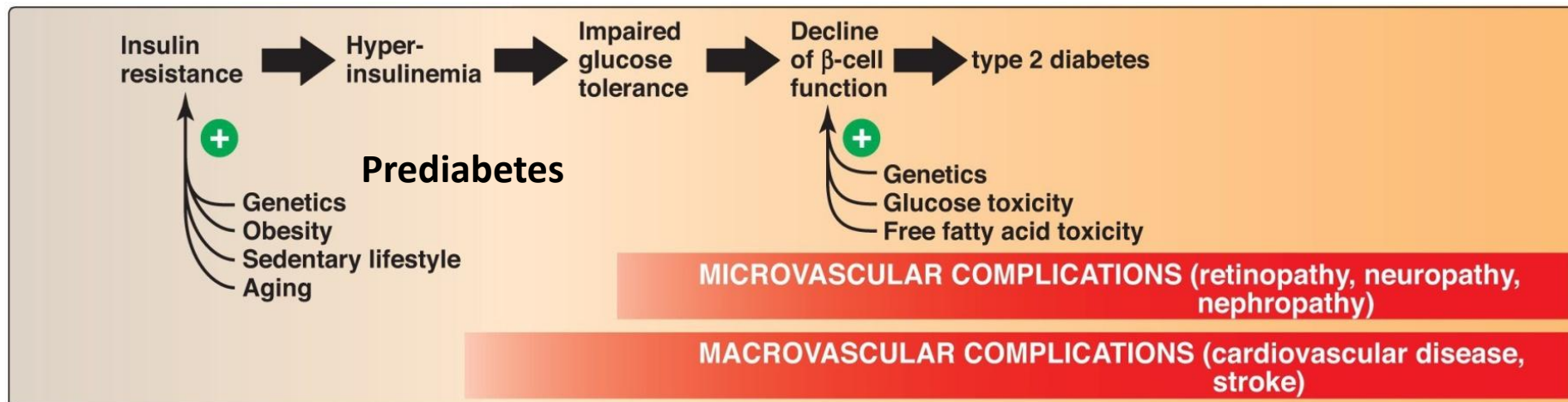
Progression of Type 2 DM and Pre-diabetes

- Pre-diabetic state (insulin resistance) develops 10-15 years before diabetes:
 - Impaired **fasting glucose** = 100-125 mg/dL → Due to **hepatic insulin resistance**
 - **2-hour Glucose tolerance test (OGTT)** = 140-199 mg/dL → Due to **skeletal muscle insulin resistance**
 - HbA1c = 5.7 – 6.4%
- Pre-diabetics higher risk of DM
- Management of pre-diabetic state:
 - Intensive diet and exercise (150 mins/week).
 - Target 7% weight loss.
- Rationale:
 - **Exercise causes insulin-independent uptake of glucose by muscle** (via AMP-kinase pathway)



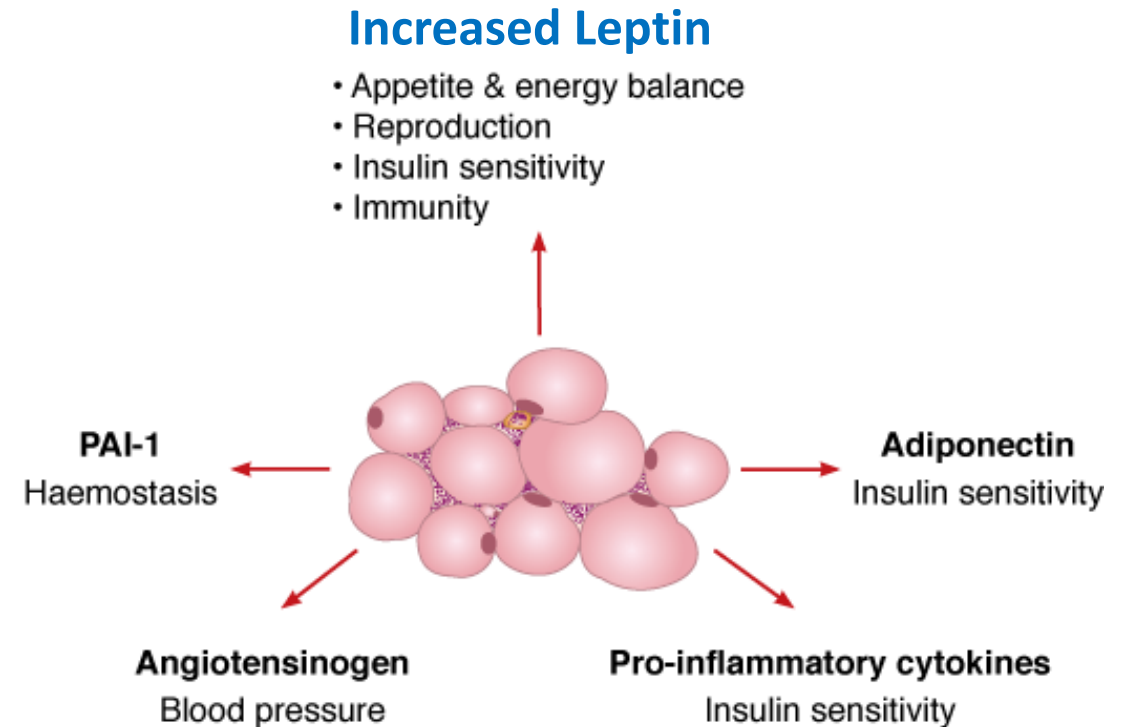
Insulin resistance

- Definition: **Less-than-expected** (suboptimal) biological effect of insulin
- A total daily **insulin dose > 200 U/day** is evidence for severe insulin resistance
- Anthropometric risk factors
 - BMI > 30 (obesity)
 - Higher waist circumference (Central obesity)
 - Waist to hip ratio (WHR) greater than 1 (Abdominal obesity)
- In central obesity, (most common cause of insulin resistance)
 - **Decreased number of insulin receptors**
 - And a **post-receptor failure**



Adipose Tissue as an Endocrine Organ

- Adiponectin:
 - Increases number of insulin responsive adipocytes
 - Stimulates fatty acid oxidation
- Leptin:
 - Synthesized and secreted in proportion to adipocyte size
 - Decreases appetite (long term)
- Weight gain (central obesity, increased adipocyte size) leads to:
 - 1. Increased leptin** secretion (due to leptin resistance)
 2. Increased pro-inflammatory cytokines
 - 3. Decreased adiponectin** secretion



Source: Gardner DG, Shoback D: *Greenspan's Basic & Clinical Endocrinology*, 9th Edition: www.accessmedicine.com
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Mechanism of Insulin Resistance

1

Obesity → **Decreased adiponectin**



Increased serum free fatty acids



Impairs insulin secretion and
impairs glucose utilization



Impairs hepatic response to insulin

2

Obesity → **Increased leptin production**



Leptin resistance → Impairs insulin action

3

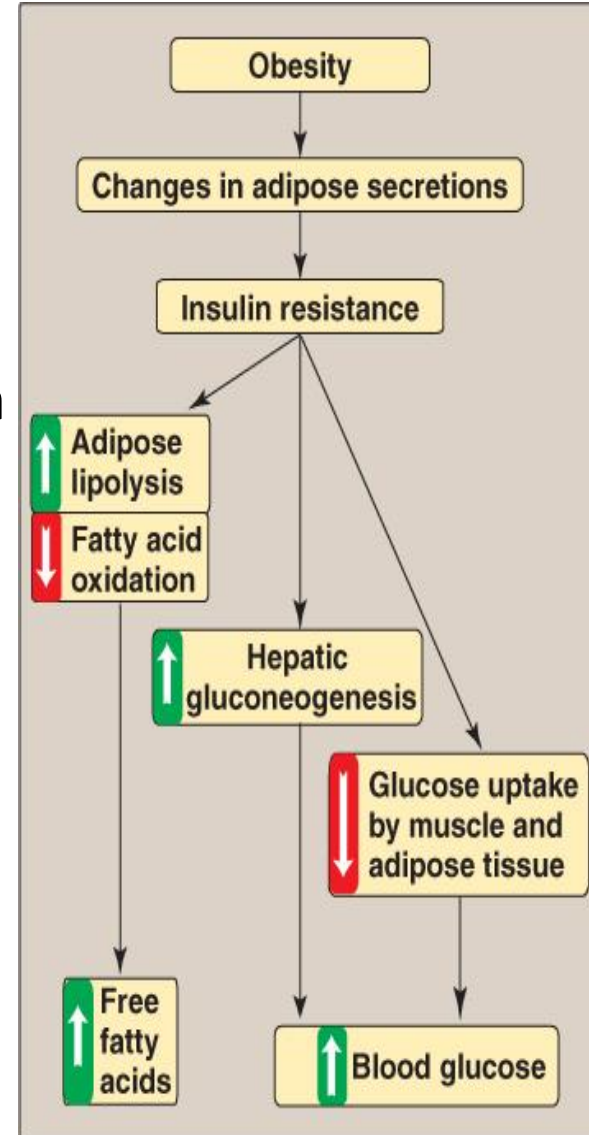
Obesity → **Decreased GLP-1** → Decreased insulin secretion
(GLP-1/incretins secreted from intestinal cells, stimulate insulin secretion)

4

Obesity → Increased proinflammatory cytokines (IL-6 & TNF- α)



Localized inflammation → Decreased insulin sensitivity



Clinical Presentation of Diabetes Mellitus

Type 1 DM	Type 2 DM
Age of onset < 15 years (commonly) Low/normal BMI Sudden onset	Age onset > 40 years (commonly) High BMI (overweight) Gradual onset (hyperglycemia for years)
Severe hyperglycemic symptoms Classic triad: polyuria, polydipsia, polyphagia Weight loss	Asymptomatic or mild hyperglycemic symptoms (found incidentally on routine blood test) Symptoms: Polyuria, polydipsia, vision changes
Diabetic ketoacidosis (acute) Microvascular and macrovascular complications (chronic) Repeated infections	Hyperosmolar hyperglycemic state (acute) Microvascular and macrovascular complications (chronic) Repeated infections
Positive islet β -cell antibodies Absent C-peptide	Negative islet β -cell antibodies C-Peptide: Normal/High/Low

Pathophysiology of Type 1 and Type 2 DM

Insulin deficiency or Insulin resistance (decreased number of insulin receptors/ post receptor defects)

Tips insulin:glucagon ratio towards glucagon

Failure to inhibit endogenous glucose production by liver

Decreased glucose uptake and glycogen synthesis in skeletal muscles after a meal

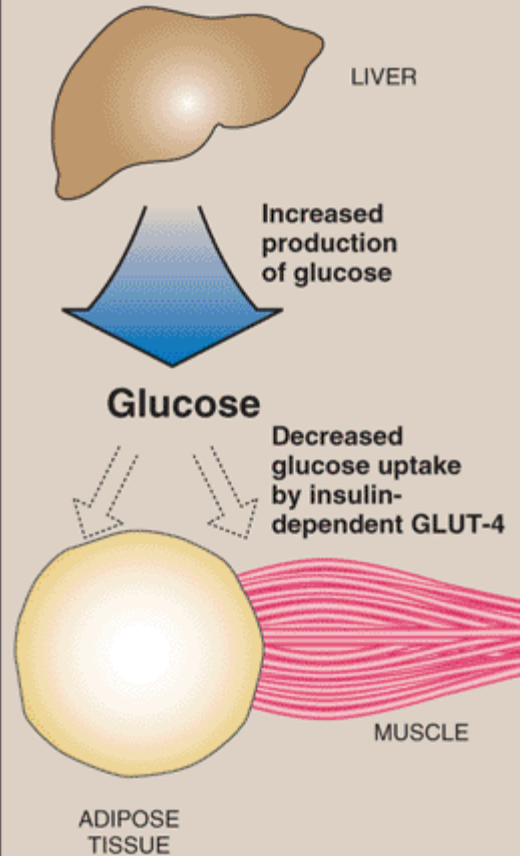
High fasting blood glucose

High postprandial blood glucose

Failure to inhibit adipose tissue lipolysis → excess triglyceride breakdown → **high circulating free fatty acids** in the blood

Increased muscle proteolysis and **negative nitrogen balance** (Insulin facilitates entry of amino acids into muscle and increases protein synthesis)

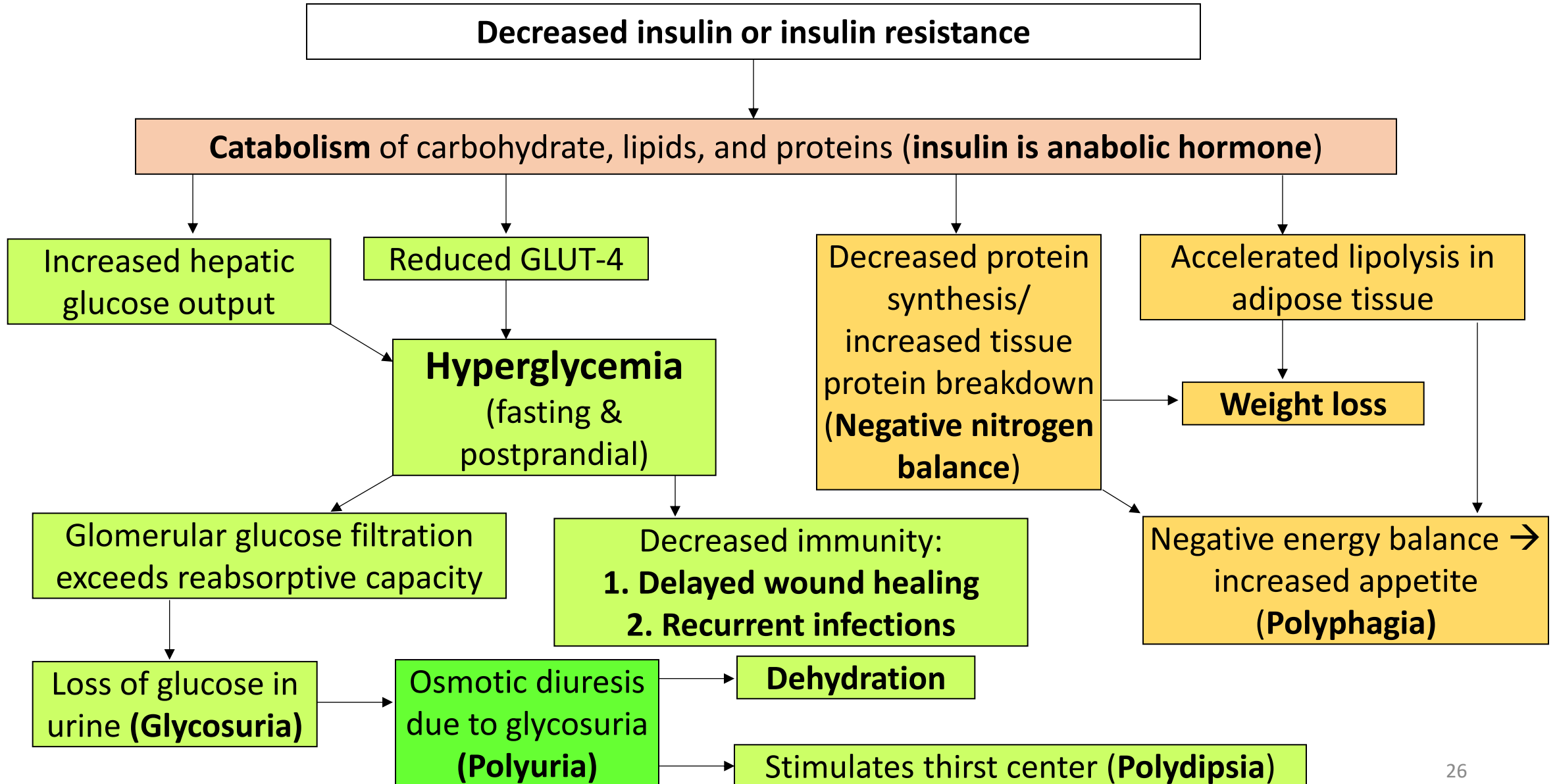
1 Insulin resistance in peripheral tissues



2 Inadequate insulin secretion from β cells



Explain signs and symptoms in DM



Diagnosis of Diabetes Mellitus (one of the criteria is required)

- Random plasma glucose ≥ 200 mg/dL with one of symptoms of diabetes (polyuria, polydipsia, polyphagia, weight loss)
- Fasting plasma glucose ≥ 126 mg/dL (repeated twice)
 - Fasting = no caloric intake ≥ 8 hours
 - Fast, sensitive, specific
- Hemoglobin A1c $\geq 6.5\%$ (repeated twice)
 - What is HbA1c? Nonenzymatically glycated hemoglobin
 - Also used to monitor blood glucose control (indicator of control over 3 – 4 months)
- 2-hour oral glucose tolerance test plasma glucose ≥ 200 mg/dL
 - Most sensitive test (considered gold standard)
 - Measure fasting plasma glucose and plasma glucose 2 hours after consumption of 75 g of glucose
 - Used in high risk situations (e.g., gestational diabetes) and to identify prediabetic state

Dyslipidemia in Insulin Resistance and Type 2 DM

- Lipoprotein abnormalities commonly observed are
 - Increased small dense LDL (LDL-B) - atherogenic
 - Decreased HDL levels
 - Increased serum triacylglycerol (Increased VLDL) – reduced activity of lipoprotein lipase – Remember, **lipoprotein lipase** requires insulin for optimum activity
 - Atherogenic lipid profile → Atherosclerosis (macrovascular) → Cardiovascular disease
- Increased circulating free fatty acids due to increased breakdown of TAGs in adipose tissue – contributes to insulin resistance and reduced insulin secretion
- Markers of insulin resistance:
 - Elevated serum free fatty acids
 - Increased serum TAG: HDL ratio

Management of Diabetes Mellitus

Essential elements in comprehensive care of Diabetes Mellitus

Individualized glycemic control:

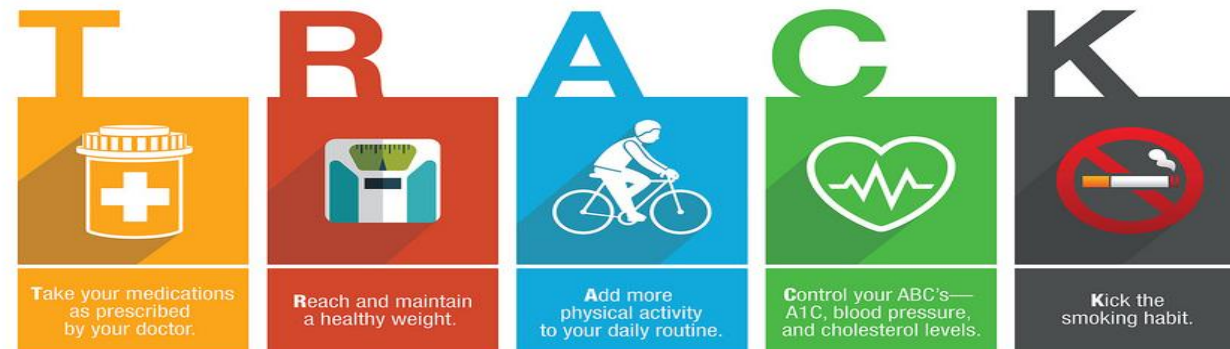
- Diet/lifestyle
- Exercise
- Medication:
 - Glucose-lowering agents
 - In Type 2 DM
 - Insulin injections
 - In Type 1 and later stages of Type 2 DM

Treat associated conditions:

- Dyslipidemia
- Hypertension
- Obesity

Screen and manage complications:

- Retinopathy
- Nephropathy
- Neuropathy
- Cardiovascular disease
- Other complications



Medical Nutrition Therapy

- **General dietary guidelines:**

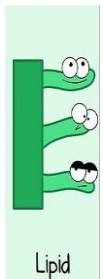
- **Higher dietary fiber** (whole grains, beans), benefits:
 1. **Decreased glycemic index** (slower glucose absorption)
 2. **Improves satiety** (reduces calorie consumption)
 3. **Reduces cholesterol absorption**
- Dietary sodium restriction if hypertensive

- **Carbohydrate in diet:**



- Food with **lower glycemic index** (slower increase in blood glucose)
- Monitor calorie intake, set limit to meals (reduce postprandial glycemia)
- **Avoid refined carbohydrate** (e.g., fructose- and sucrose-containing beverages), replace with nonnutritive sweetener (non-caloric)

- **Fat in diet** (targeting dyslipidemia, reducing cardiovascular risk):



- **Low-saturated fatty acids**
- No trans-fat consumption
- Reduction in dietary cholesterol (200 – 300 mg/day)
- **Lipid content 25 – 30% of total caloric consumption**
- Increase omega-3: omega-6 ratio (improves cardiovascular health)

Management of Diabetes Mellitus

■ Physical Activity:

- Improves insulin sensitivity, how?
 - i. Increases metabolism and lean body mass
 - ii. Increases GLUT-4 and glucose uptake
 - iii. Reduces visceral (central) obesity
 - iv. Decreases serum TAGs and improves HDL
- Reduces cardiovascular risk and blood pressure
- Maintains muscle mass, reduces body fat, improves weight loss
- Recommendation is 150 min/week (over at least 3 days) of moderate aerobic physical activity with no gaps > 3 days
- Type 1 DM patients should monitor blood glucose before, during, and after exercise to avoid exercise-related hypoglycemia

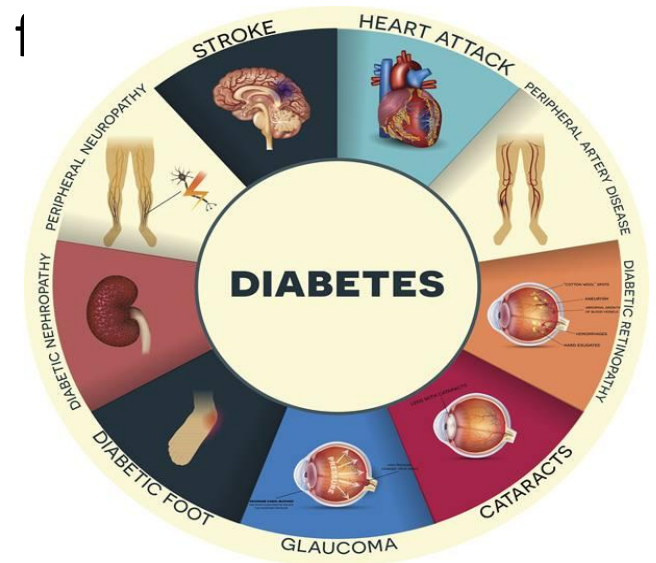
■ Other lifestyle modifications:

- Smoking cessation
- Weight reduction if overweight/obese, maintain BMI ideally at 22 – 25

Management of Diabetes Mellitus

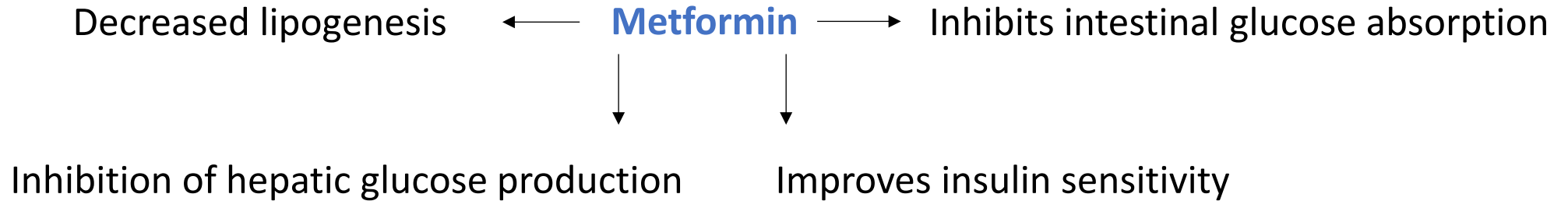
■ Health Maintenance:

- Glycemic control (check HbA1c every 3 – 6 months)
- Monitor for microvascular complications (performed annually):
 - Retinopathy (eye exam)
 - Nephropathy, check for microalbuminuria (urine albumin/creatinine ratio)
 - Peripheral neuropathy, diabetic foot (comprehensive 1
- Reduce cardiovascular risk:
 - Lifestyle modifications
 - Annual lipid and blood pressure screening
 - Aspirin and statin (depending on other risk factors)

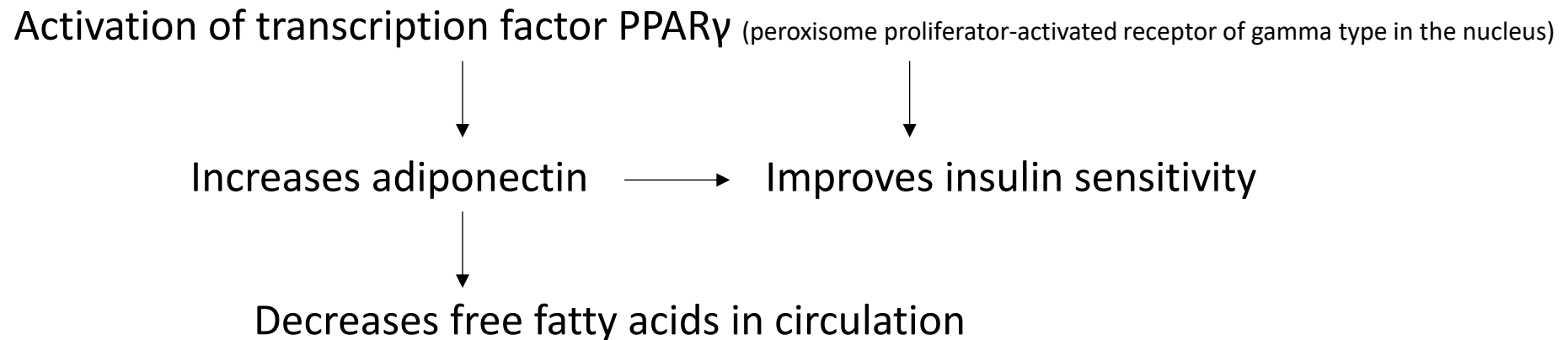


Management of DM – Pharmacologic Basis

Drugs that improve insulin sensitivity



TZDs (Thiazolidinediones, glitazones)



Management of DM – Pharmacologic Basis

Drugs that improve insulin secretion

Sulfonylureas: Blocks ATP-sensitive potassium channels → insulin secretion

GLP-1 agonists:

1. Enhances glucose-mediated insulin release
2. Slows gastric emptying and promotes satiety
3. Decreases glucagon secretion

Other glucose-lowering agents

α -glucosidase inhibitors: Decreases dietary carbohydrate absorption

SGLT-2 inhibitors: Inhibition of SGLT-2 (Sodium-glucose cotransporter) in proximal tubule of the kidney → decreased glucose reabsorption in the kidney

Management of DM – Pharmacologic Basis

Insulin Therapy

- Recombinant human insulin
- Type 1 diabetes mellitus: life-long insulin supplements
- Type 2 diabetes mellitus: in later stages, during stress (e.g., elective surgery, infections)
- Standard vs intensive insulin therapy
- Intensive insulin therapy – Rigorous control of plasma glucose –better HbA1c and lower risk of complications.
 - But, high risk of hypoglycemia due to insulin overdose
- Standard insulin therapy – less rigorous control of plasma glucose and slightly higher HbA1c levels – higher risk of long-term complications.
 - But, lower risk of hypoglycemic episodes
- Recognize hypoglycemia and immediate management

Blood Glucose and Hypoglycemia

Hypoglycemia

Normal blood glucose ranges

Fasting: 70 – 100 mg/dL (3.9 – 5.6 mmol/L)

2 hours postprandial: < 120 mg/dL (< 6.6 mmol/L)

Prediabetic: 100 – 125 mg/dL (3.9 – 6.9 mmol/L)

Hypoglycemia Definition

- In non-diabetics: blood glucose level < 55 mg/dL (< 3.0 mmol/L) with adrenergic or neuroglycopenic symptoms that resolves after glucose levels are raised
- In diabetics: < 70 mg/dL (higher threshold to alarm patient to take action)

Glucose has a special role in metabolic homeostasis

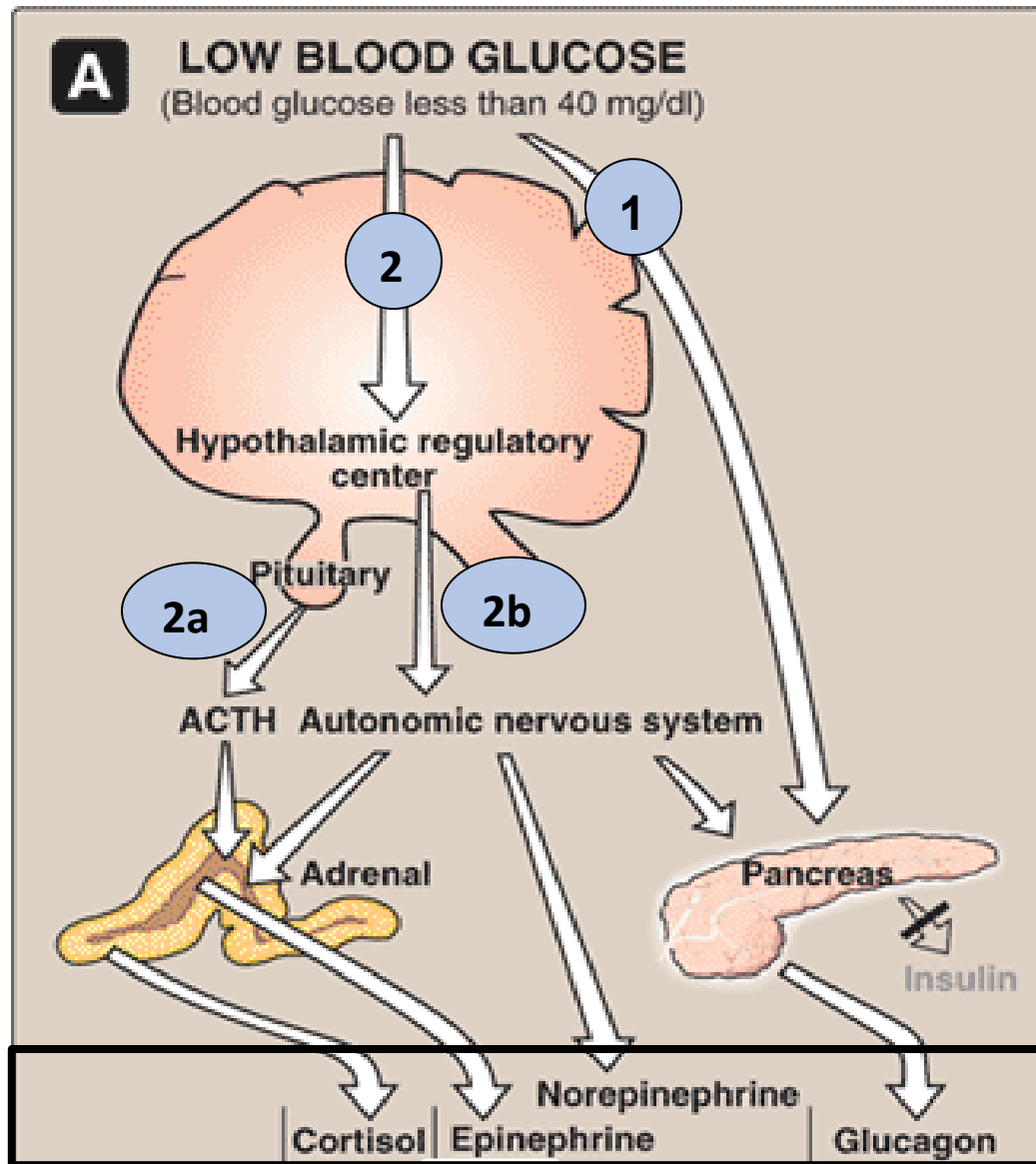
**Many cells depend on glucose for their metabolism
(RBC, brain, retina as examples)**

- **Insulin** and **glucagon** are the **main hormonal regulators** of blood **glucose** levels and the release of insulin or glucagon responds directly to glucose in the blood
- **Glucagon, epinephrine, cortisol** and **growth hormone** are often grouped as **insulin counterregulatory hormones** and are released to prevent **hypoglycemia**

Low blood glucose leads to release of insulin counterregulatory hormones

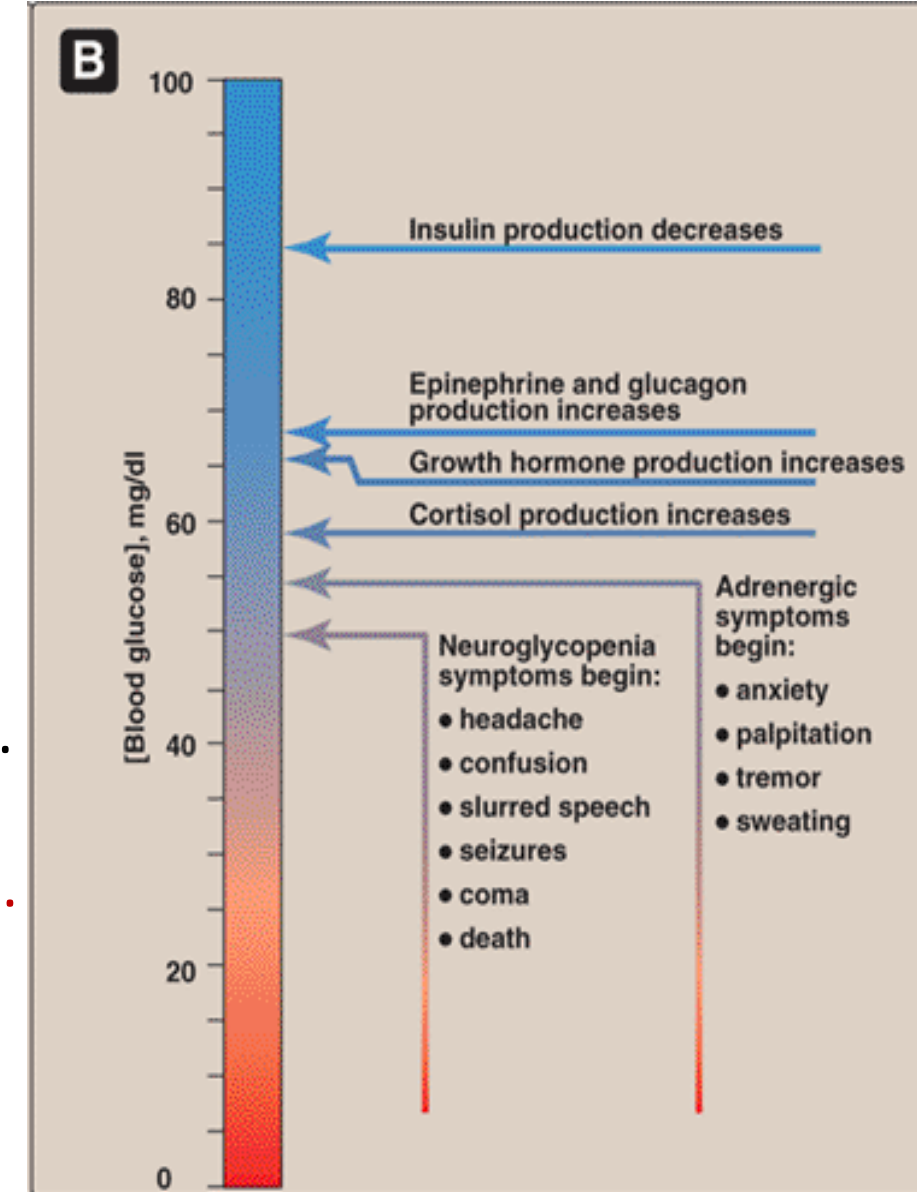
These hormones are released by two overlapping glucose-regulatory systems:

- 1 Low glucose → low insulin → **glucagon** released by α -cells → hepatic gluconeogenesis & glycogenolysis
- 2 Hypothalamic glucoreceptors can trigger:
 - 2a Anterior pituitary releases **ACTH** & **growth hormone**.
ACTH → **cortisol** release → **epinephrine** secretion (adrenal medulla):
Hepatic gluconeogenesis & glycogenolysis, & mobilization of gluconeogenic substrates
 - 2b Secretion of **catecholamines** through the **sympathetic** nervous system



Response to low blood glucose levels

- Less than 70 mg/dL → Increased release of:
 1. **Glucagon** (2nd line of defense)
 2. **Epinephrine** (3rd line of defense)
 3. **Growth hormone** (prolonged hypoglycemia beyond ~4 h)
 4. **Cortisol** (When hypoglycemia is prolonged beyond ~4 h)
- Less than 55 mg/dL:
 - Onset of **adrenergic** and **neuroglycopenia** symptoms.
- Transient **hypoglycemia** can lead to **cerebral dysfunction**.
- Severe **prolonged hypoglycemia** can cause **coma** and **death**.



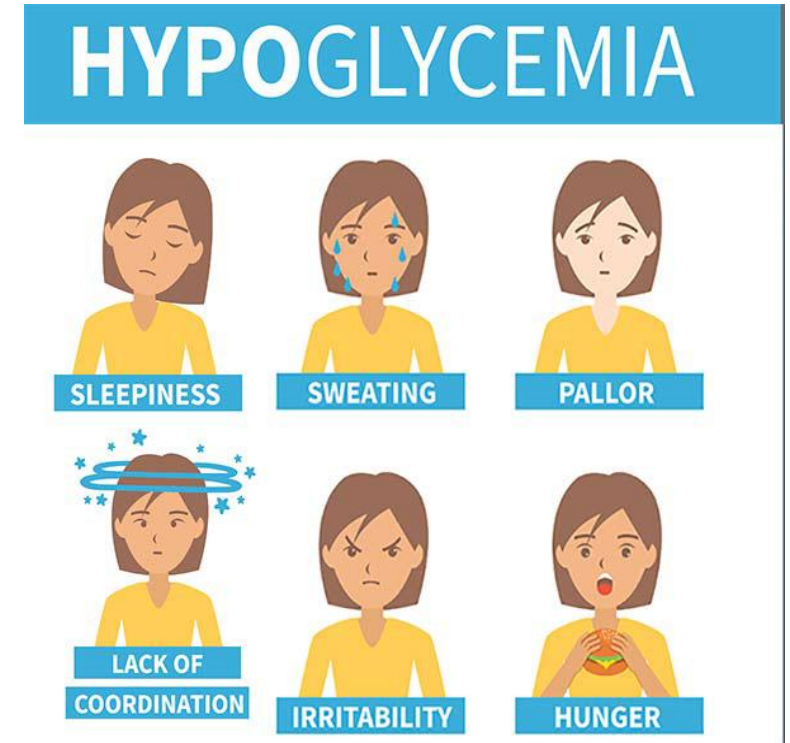
Adrenergic or Neuroglycopenia Symptoms

- **Adrenergic symptoms:**

- **Abrupt decline** in blood glucose → hypoglycemia triggers sympathoadrenal discharge → **epinephrine release**
- **Palpitations, anxiety, tremors, sweating, tachycardia**

- **Neuroglycopenic symptoms:**

- **Gradual decline** in blood glucose
- Results directly from **CNS glucose deprivation** (**fails to trigger epinephrine response**)
- Behavioral changes, headache, confusion, slurred speech → coma → death



Causes of hypoglycemia

- **Diabetics:**
 - Insulin excess (accidental overdose, drug interactions), fasting, exercise
- **Endogenous hyperinsulinism, example:**
 - **Insulinoma** (most common)
- **Exogenous hyperinsulinism, example:**
 - **Factitious disorder**
- **Hormone deficiencies, example:**
 - Adrenal insufficiency
- **Genetic and congenital disorders, examples:**
 - Glycogen storage diseases, fructose intolerance
- **Others:**
 - **Alcohol-related fasting hypoglycemia**
 - Reactive postprandial hypoglycemia (transient)
 - Critical illness



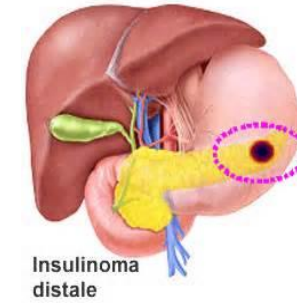
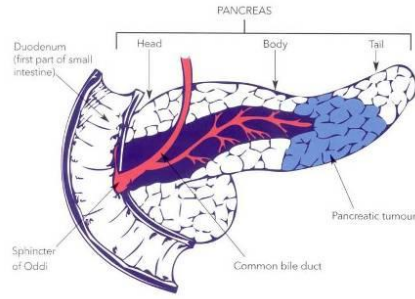
Alcohol-related fasting hypoglycemia

- **Common in undernourished, longer fasting or dehydrated individuals after consumption of many drinks with high alcohol content.**



- **Alcohol metabolism in liver** uses enzymes that convert NAD^+ to NADH
- **Ethanol metabolism $\rightarrow$ Abnormal high NADH/NAD^+ ratio in hepatocytes $\rightarrow$ Reduces gluconeogenesis**
 - Most individuals experience mild and transient fasting hypoglycemia which improves after food intake.
- **Chronic intake of alcohol may also lead to fasting hypoglycemia.**

Hypoglycemia due to insulinoma



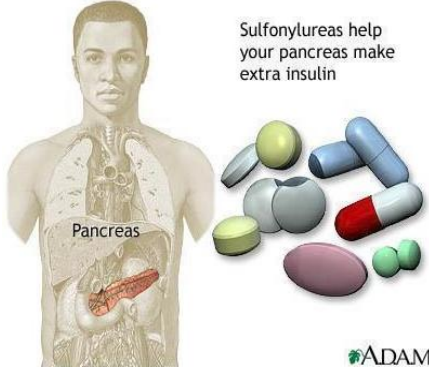
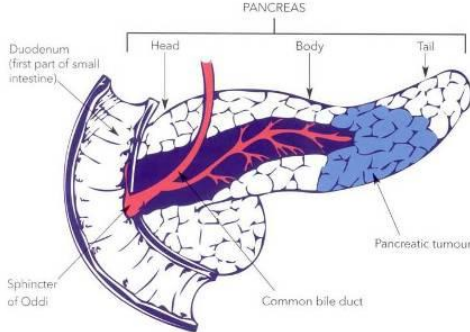
- **Insulinoma: Tumor of the pancreatic islet cells that releases insulin.**
- **Severe hypoglycemia** in patients with **insulinoma** due to excessive **uptake** of blood **glucose** into **skeletal muscle** and **fat cells** by GLUT-4.
- **Abnormal high** level of serum **insulin** **blocks** action of the **insulin counter-regulatory hormones**.
- The **blood** contains **high** levels of **insulin** and **C-peptide** and **elevated** amounts of **proinsulin**

Factitious hypoglycemia

- **Factitious disorders** can be found in patients who **simulate** a **disease** or artificially **induce clinical symptoms**.
- For example, a **non-diabetic** individual with factitious disorder may obtain drugs for treatment meant for a diabetic patient and **induces factitious hypoglycemia** by using:
 1. **Insulin** injection
 2. **Sulfonylurea** tablets

Differentiate: Measure insulin, C-peptide, proinsulin and sulfonylurea.

Differentiate hypoglycemia in a non-diabetic patient

Blood Levels	Injection of Insulin	Oral Sulfonylurea	Insulinoma
			
Insulin	High	High	High
C- Peptide	Low ★	High	High
Proinsulin	Low to absent ★	Present	High
Sulfonylurea	Absent	Present ★	Absent

Hypoglycemia Management

Treatment

Conscious patient: Oral consumption of glucose.



Unconscious patient: Injection of glucagon or epinephrine to activate hepatic glycogen degradation and release of glucose into the blood.
Intravenous dextrose/ glucose



Summary: Hypoglycemias and Hyperglycemias

■ Hypoglycemias:

1. **Hypoglycemia due to insulinoma** (Insulin secreting tumor of pancreas)
2. **Hypoglycemia due to high insulin**
 - Insulin injection (high insulin)
 - Exaggerated postprandial response (high insulin)
 - Factitious disorder (high insulin from insulin injection or sulfonylurea).
3. **Hypoglycemia due to reduced gluconeogenesis**
 - Hereditary disorders
 - Alcohol induced during fasting hypoglycemia.

■ Hyperglycemias:

1. **Under secretion of insulin**
 - **Type 1 Diabetes Mellitus:** Atrophy or destruction of β -cells of pancreas due to immune response or viral infection.
2. **Decreased sensitivity to insulin**
 - **Type 2 Diabetes Mellitus:** Insulin resistance - more insulin than normal is needed for receptors to respond.